Internship Opportunity – Improving DAS Signal Quality Through Signal Processing and Artificial Intelligence

Background

Distributed Acoustic Sensing (DAS) turns an optical fibre into a continuous array of acoustic sensors capable of measuring vibrations over several kilometres. Its operation relies on Rayleigh backscattering: when a light pulse travels through the fibre, a small fraction of the light is scattered back towards the source due to microscopic irregularities in the glass. The phase and amplitude of the backscattered light change when the fibre is strained, making it possible to detect and locate vibrational events.

The SURFO team uses DAS to monitor railway infrastructure in real time. Applications include train traffic monitoring, detection of track and rolling-stock defects, identification of external events such as rockfalls, and intrusion detection alongside railway tracks. A single optical fibre thus acts as a dense array of thousands of distributed sensors.

Challenge

Analysing DAS data presents several technical challenges. Noise reduction is often essential to obtain usable information under complex measurement conditions. High-amplitude events may cause signal saturation, while the considerable volume of generated data calls for suitable compression and processing techniques to enable efficient use of the measurements.

Internship objective

The aim of this internship is to develop signal-processing and/or artificial-intelligence algorithms that significantly improve the quality of data acquired by a DAS system.

Internship tasks

The work will focus on several complementary topics:

- **Saturation detection and correction**: develop methods to identify and correct saturated signals in order to preserve information during high-intensity events.
- **Noise reduction**: design denoising algorithms to improve the signal-to-noise ratio, particularly through deep-learning approaches.
- **Fast demodulation**: optimise demodulation methods to enable real-time processing.
- **Data compression**: develop compression methods suited to the characteristics of DAS signals while preserving information relevant to event detection.

The first two topics will be prioritised. The remaining two may be explored depending on progress and the intern's interests.

The internship will follow several successive stages:

- Become familiar with the available DAS data and carry out an in-depth analysis;
- Review relevant signal-processing and/or AI approaches in the literature;
- Design, implement and evaluate innovative algorithms;
- Validate the methods on real-world data and analyse their performance.

Candidate profile

Education

Final-year engineering student or second-year Master's student specialising in signal processing, artificial intelligence or applied mathematics.

Technical skills

- Strong foundations in mathematics, signal processing and/or *machine learning*;
- Proficiency in Python (NumPy, SciPy and PyTorch are appreciated);
- Ability to analyse experimental data.

Personal qualities

- Scientific rigour, curiosity and independence;
- Teamwork and good communication skills.

Practical information

- Duration: 5 to 6 months
- Location: La Plaine-Saint-Denis, France
- Start date: from 1 March 2027
- Supervision: Mahran RIHANI, Imen BEN AMARA
- Allowance: in accordance with applicable regulations
- Benefits: SNCF travel benefits.

How to apply

Please send your CV and cover letter to all the following addresses:

- `ext.mahran.rihani@reseau.sncf.fr`
- `ext.imen.benamara@reseau.sncf.fr`
- `ext.tarik.hammi@reseau.sncf.fr`
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References

- Chen-Ray Lin et al., *Analysis of saturation effects of distributed acoustic sensing and detection on signal clipping for strong motions*, Geophys. J. Int. (2025) 241, 971–985.
- Fan Cunzheng et al., *Large Dynamic Range Optical Fiber Distributed Acoustic Sensing (DAS) With Differential Unwrapping-Integral Algorithm*, Journal of Lightwave Technology, 2021.
- Zhixiang Wei et al., *An improved phase unwrapping method based on deep learning for optical fiber sensors*, Optics Communications, Volume 598, 2026.
- Kowarik Stefan et al., *Fiber optic train monitoring with distributed acoustic sensing: Conventional and neural network data analysis*, Sensors 20(2), 2020.